

A working example

Here is an example of RSA encryption and decryption. The parameters used here are artificially small, but you can also use OpenSSL to generate and examine a real keypair.

1. Choose two random prime numbers.
2. : $p = 61$ and $q = 53$; Compute $n = pq$
3. : $n = 61 * 53 = 3233$
4. Compute the totient $\phi(n) = (p - 1)(q - 1)$
5. : $\phi(n) = (61 - 1)(53 - 1) = 3120$
6. Choose $e > 1$ coprime to 3120
7. : $e = 17$
8. Choose d to satisfy $de \bmod \phi(n) \equiv 1$
9. : $d = 2753$
10. : $17 * 2753 = 46801 = 1 + 15 * 3120$.

The **public key** is ($n = 3233$, $e = 17$). For a padded message m the encryption function $c = m^e \bmod n$ becomes:

$$c = m^{17} \bmod 3233$$

The **private key** is ($n = 3233$, $d = 2753$). The decryption function $m = c^d \bmod n$ becomes:

$$m = c^{2753} \bmod 3233$$

For example, to encrypt $m = 123$, we calculate

$$c = 123^{17} \bmod 3233 = 855$$

To decrypt $c = 855$, we calculate

$$m = 855^{2753} \bmod 3233 = 123$$

Both of these calculations can be computed efficiently using the square-and-multiply algorithm for modular exponentiation.